Roma Tre

Information Visualization

Design Models

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Overview

- The target visualization system
  - Rhetoric and dialectic paradigms
- Design models
  - Nested model
  - Linearized model
  - Some examples

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Credits

- These slides were realized in collaboration with Giuseppe Liotta and Alessandra Tappini from the University of Perugia
- Some slides are based on those produced by Tamara Munzner (UBC University)
  - the slides by Munzner are publicly available at https://www.students.cs.ubc.ca/~cs-447/23Jan/

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The target visualization system

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Rhetoric

Rhetoric

/ˈrɛʈərɪk/

noun [uncountable]

from the Greek ῥητορική (τέχνη)

The skill of using language in speech or writing in a special way that influences or entertains people.

Oxford Languages

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Dialectic

Dialectic

/ˌdaɪəˈlɛktɪk/

noun [singular]

from the Greek διαλεκτική (τέχνη)

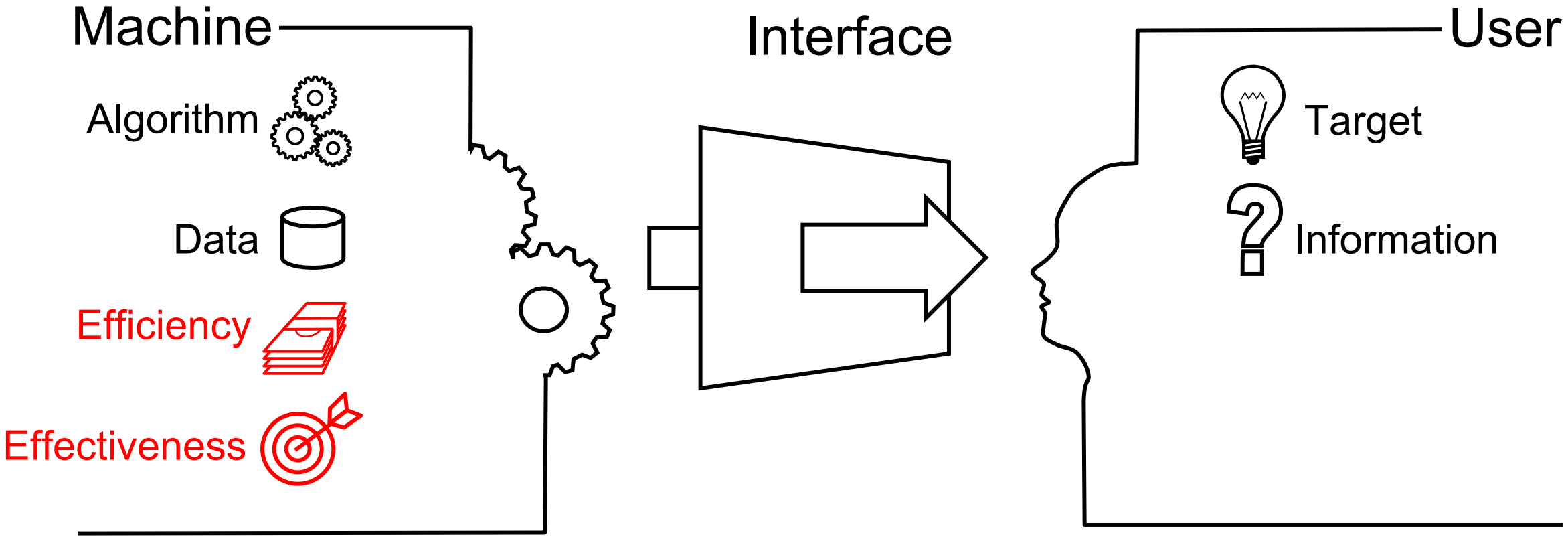
A method of discovering the truth of ideas by discussion and logical argument, and by considering ideas that are opposed to each other.

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Information Visualization

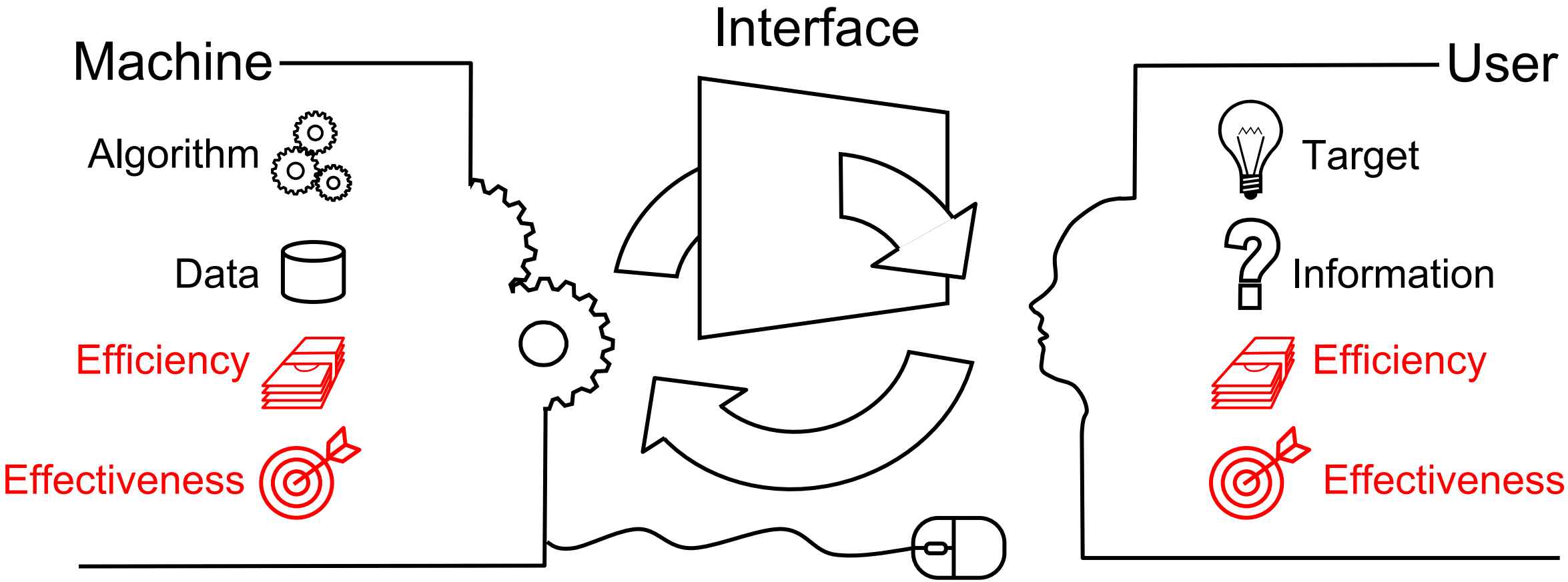
- Old-fashioned (rhetoric) user-machine interaction paradigm



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Information Visualization

- Current (dialectic) user-machine interaction paradigm



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Comparison between visualization paradigms

	Rhetoric	Dialectic
user role	passive	active
quality	of the machine	of the interaction
system target	produce readable representations	help the user carrying out tasks
systems	general purpose	specific for a task

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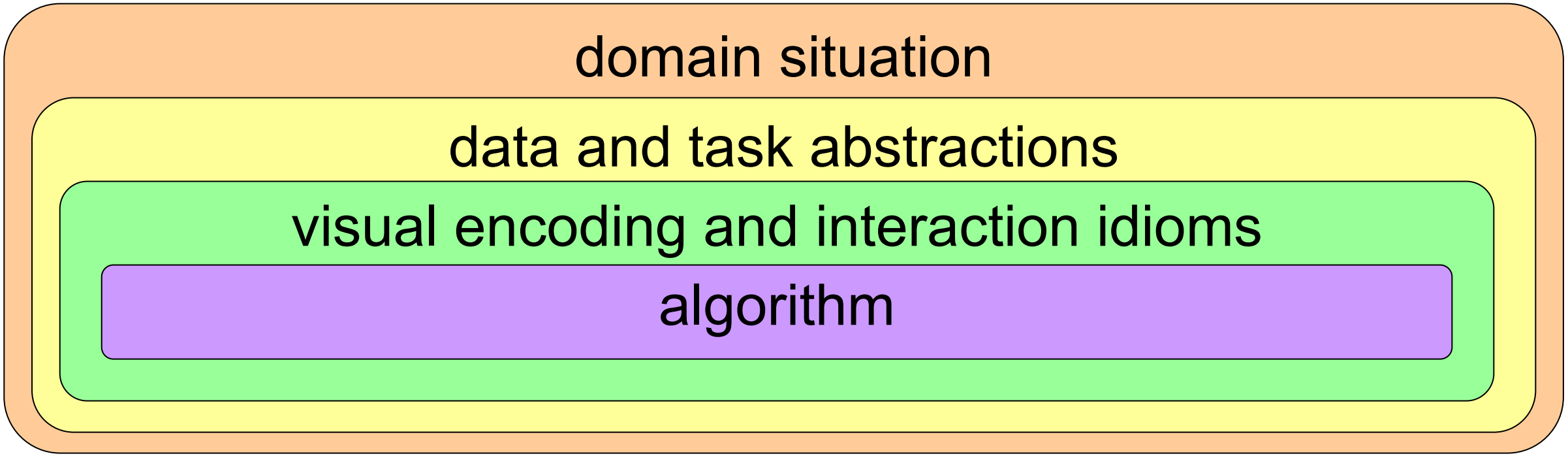
The nested model

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Munzner's nested model



- Munzner proposes a task-oriented model of infovis design composed of four activities
  - From [Munzner '14]



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1. Domain situation

- Identify the specific problem that you want to solve
- Describe a specific situation, that is
  - a group of target users
  - their domain of interest
  - their questions
  - their data
- The outcome of the design step is the understanding of the needs of the user

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2. Data and task abstraction

- Model the abstract data that you have to visualize
  - data may need to be transformed from the available domain model
- Model the abstract tasks that the user performs
  - examples of abstract tasks
    - sorting, filtering, characterizing trends and distributions, finding anomalies and outliers, and finding correlation, etc.
- The outcome of this design step is a model

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3. Visual and interaction idioms

- Visual encoding idiom
  - decide how to create pictures from the data
  - choose among the various alternative visualizations
- Interaction idiom
  - decide how the interface reacts to the user actions
  - decide how the users change what they see

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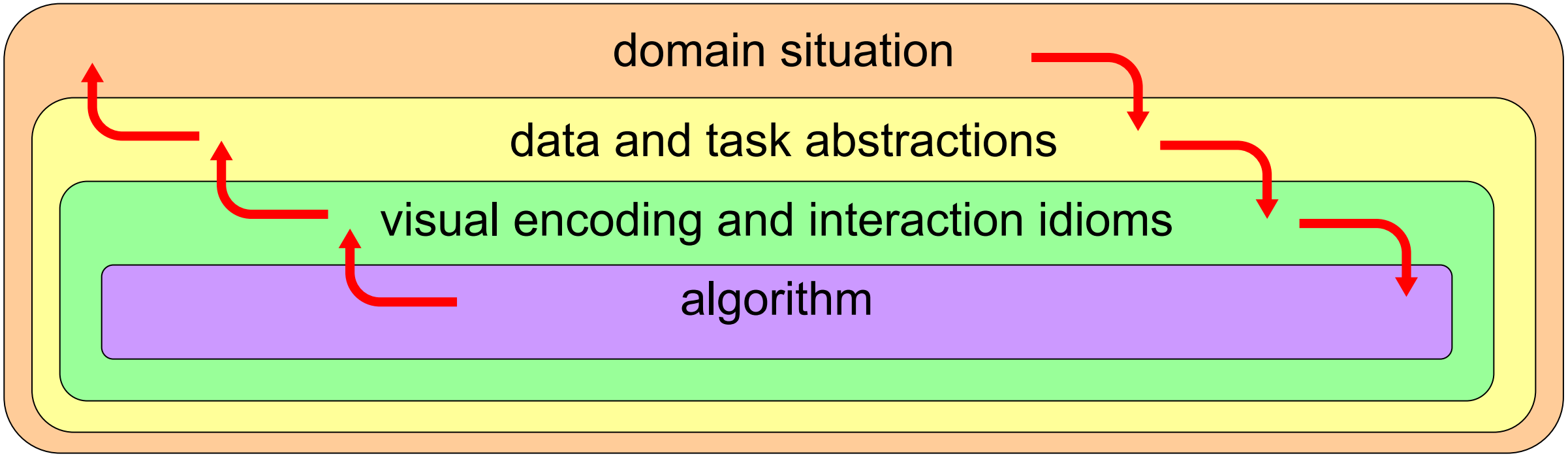
4. Algorithm

- Devise a detailed procedure that allows a computer to automatically carry out a desired goal
  - different algorithms may correspond to the same idioms

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Bottom-up and top-down traversals

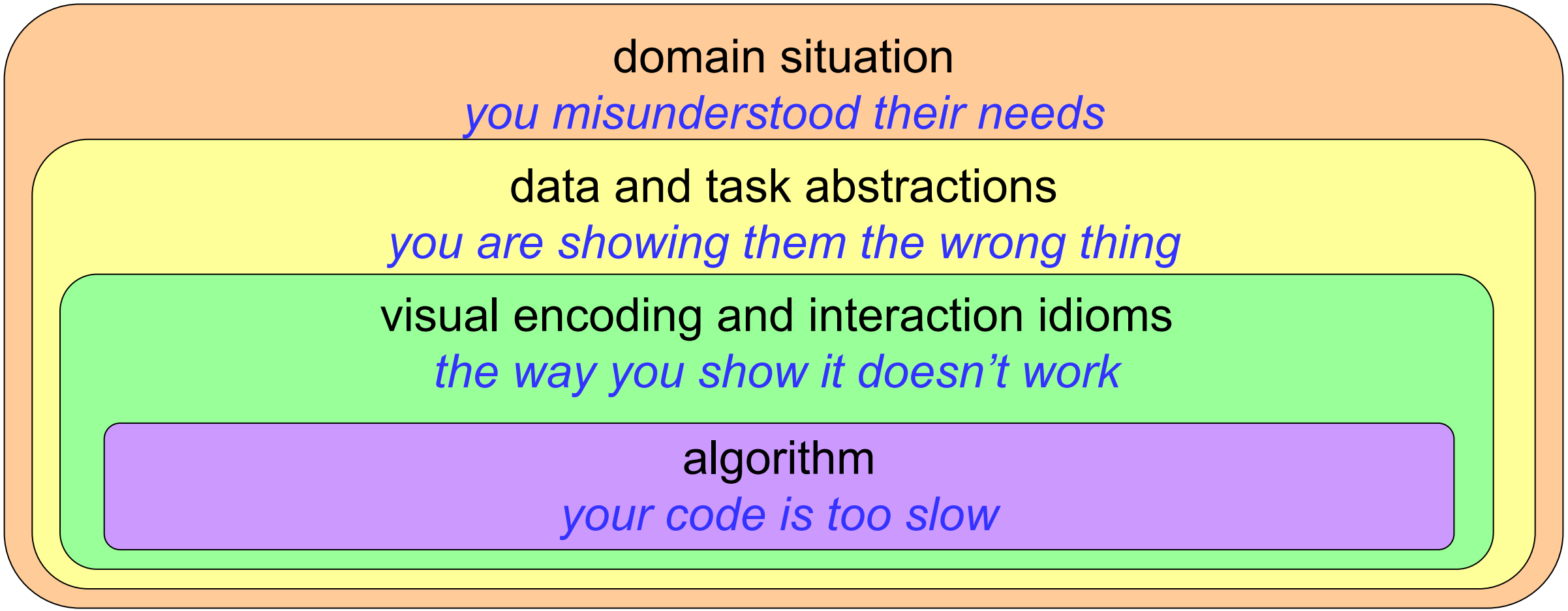
- Downstream: cascading effects
- Upstream: iterative refinement



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Errors and design choices

- Different ways to get it wrong at each level



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Validation of a project

- Involves methods from different fields at each level
  - Domain situation (anthropology, ethnography)
    - observe target users using existing tools
    - measure adoption
  - Data and task modeling (anthropology, ethnography)
    - observe target users after deployment (field study)
  - Visual encoding and interaction idioms (cognitive psychology)
    - justify design with respect to alternatives
    - analyze results qualitatively
    - measure human time with lab experiments (lab study)
  - Algorithm (computer science)
    - measure system time and memory usage
    - analyze computational complexity

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The linearized model

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Why the linearized model?

- It enriches the nested model with some important design phases (architectural and technological choices, realization and deployment) offering a more complete design process
- It is closer to traditional software engineering point of view for system design

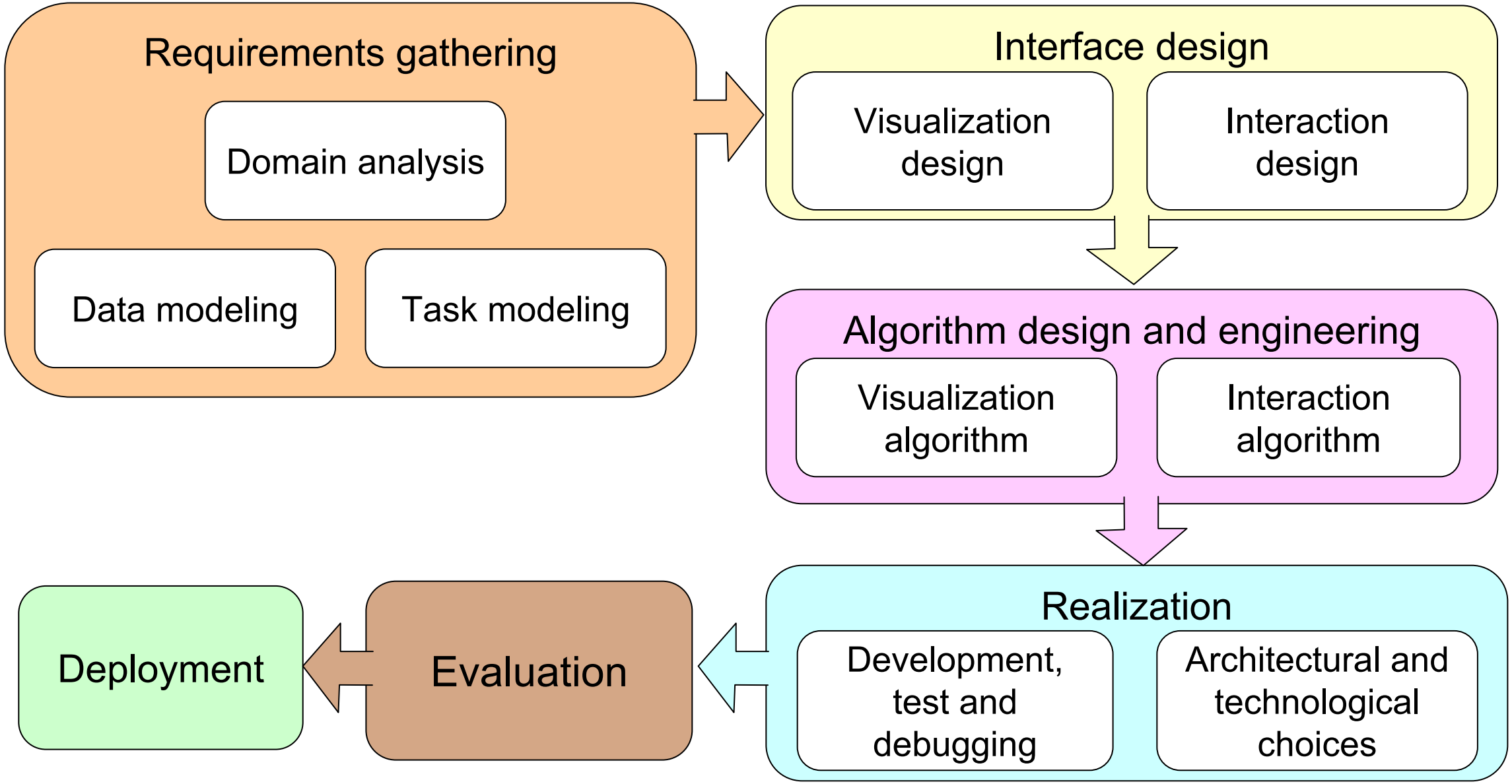
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A metodology for infovis projects

- An infovis project is a special case of a software engineering project
  - its purpose is to allow the users to interact with a graphical interface in order to perform their tasks
  - some phases of the project are shared with traditional software projects

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Design pipeline



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Requirements gathering

- Purposes of requirements gathering
  - identify the relevant stakeholders
  - establish project goals and objectives
    - equivalent to Munzner's "domain situation" phase

Traditional software projects	Infovis projects
The process pipeline is clearly defined and only needs to be explicited	The process pipeline is unknown and cannot be fully described ▪ no use-case available
The kind of data/information involved in the process is well known	The part of the data that provides information is not known a priori

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Restricting the targets

- Focus on a restricted set of users and on a restricted set of decisions that the users need to take
  - ideally a single kind of user and a single kind of decision
  - a decision may coincide with
    - an action that the user needs to take
    - a statement that user needs to assert
    - an hypothesis that the user needs to prove or to falsify
    - ...

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Examples of users and goals

- Suppose your data are tweets from Twitter
  - for each tweet you have: text, user, timestamp, location, ...
- Possible stakeholders and objectives may be
  - users who want to see the top hashtags currently mentioned by Twitter users in Italy
  - Twitter users who want to assess the activities of those people they are following
  - company managers who want to monitor some hashtags related to their company and the general sentiment associated with them
  - researchers who want to undestand the differences among tweets from US and Canada
  - …

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Task modeling

- Equivalent to Munzner's "task abstraction" phase
- An *elementary task* is a simple operation that can be a part of a more complex one
  - for example: compare two quantities
- Purpose of task modeling
  - identify the elementary tasks that contribute to the decision process
    - the decision process itself may remain unknown
  - list and prioritize the elementary tasks
    - neglect elementary tasks that are not needed in order to take the target decisions

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Examples of task modeling

- Suppose your data are tweets from Twitter
- Possible elementary tasks could be
  - focus on the last minute/hour/day
  - determine a clear ranking of all the hashtags
  - restrict to the top ten hashtags
  - identify the time-windows when a user is active
  - consider the amount of tweets from a specific geographic location
  - ...

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Data modeling

- Purpose of data modeling
  - formally describe the data that is needed to perform the elementary tasks
- Data modeling is actually a synthesis process starting from the available data
  - data curation = data extraction, data cleaning, data integration, data maintenance...

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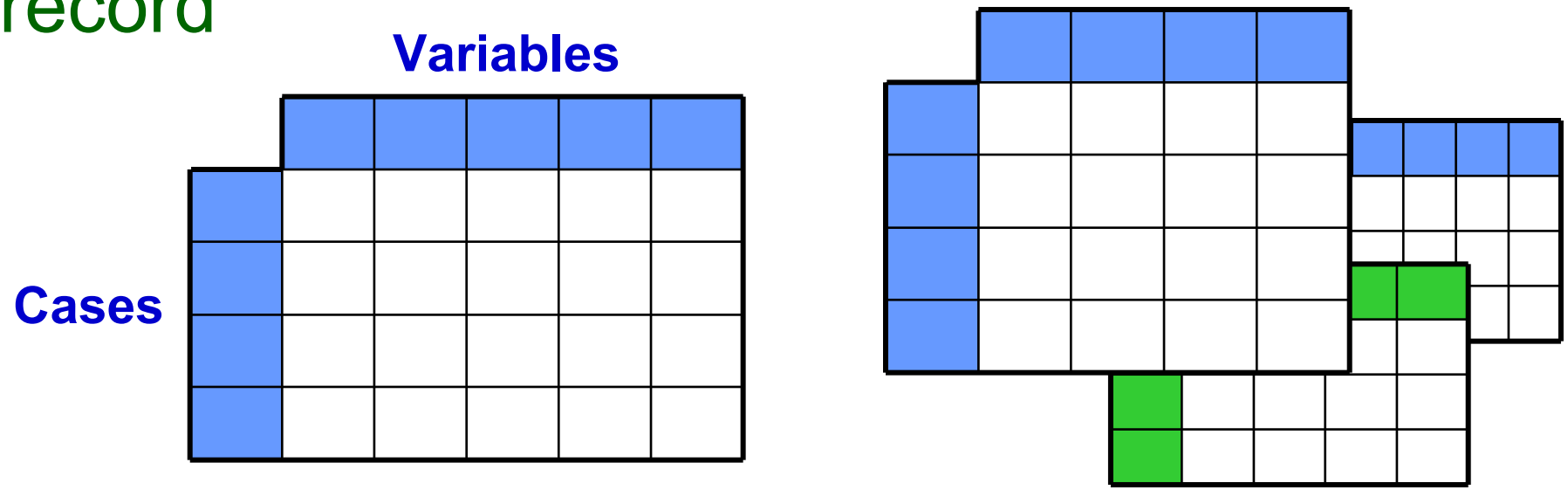
Examples of data modeling choices

- Suppose your data are tweets from Twitter
- Possible data modeling choices include
  - parse the tweet texts in order to extract hashtags
  - filter out all tweets that do not have Italy as geographic location
  - project the data removing the text and the user information
  - ...

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Data modeling output: data tables

- Data is transformed into a model that is more workable
  - single entities are called *cases*, or *records* or *data points*
  - attributes are called *variables* and have a value for each record



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Visualization design

- Purpose of visualization design
  - decide how the interface is organized into different views and windows
  - decide the *visual idiom*
    - decide how to represent data cases and variables by means of geometric objects (aka *marks*)
    - decide how colors, shapes, sizes and lengths are used to encode meaningful information (aka *channels*)
- The output of this activity is a *visual language*
  - meant to describe the main concepts at hand
  - it may use several visual idioms at once

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Example of visualization choices

- Suppose your data are tweets from X (Twitter)
- Possible visualization choices include
  - associate each hashtag with a graphic object whose size represents the number of tweets
  - animate the real-time arrivals of tweets containing specific hashtags
  - give a positional representation of the hashtags ranking
  - ...

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Interaction design

- Purpose of interaction design
  - decide how the system reacts to user actions
  - decide how the user can filter, project, select data
  - decide how actions taken on different views are related to each other
- The output of this activity is a dialectic language
  - meant to extract meaning from the data via a sequence of actions (questions) and reactions (answers)

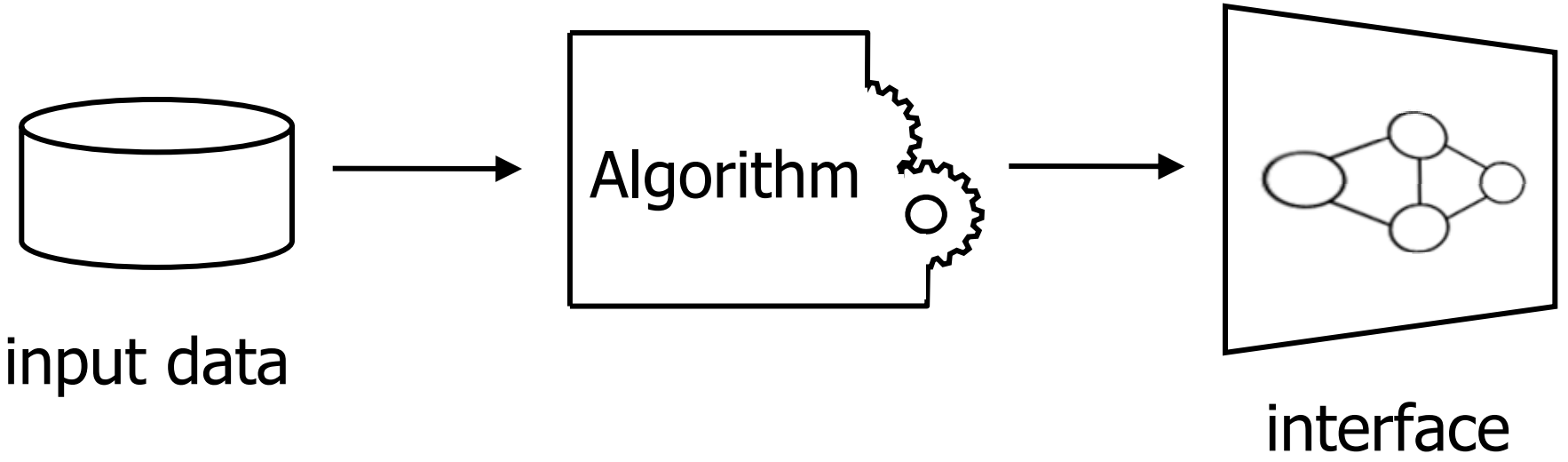
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Example of interaction design choices

- Suppose your data are tweets from X (Twitter)
- Possible interaction design choices include
  - use the scrollbar to see the lower part of the hashtag ranking
  - use a button to filter-out retweets
  - use a button to swap between linear or logarithmic representations of the quantities at hand
  - use a pop-up on hashtags to see relevant statistic values
  - select a portion of a map to restrict the geographic location of the tweets
  - ...

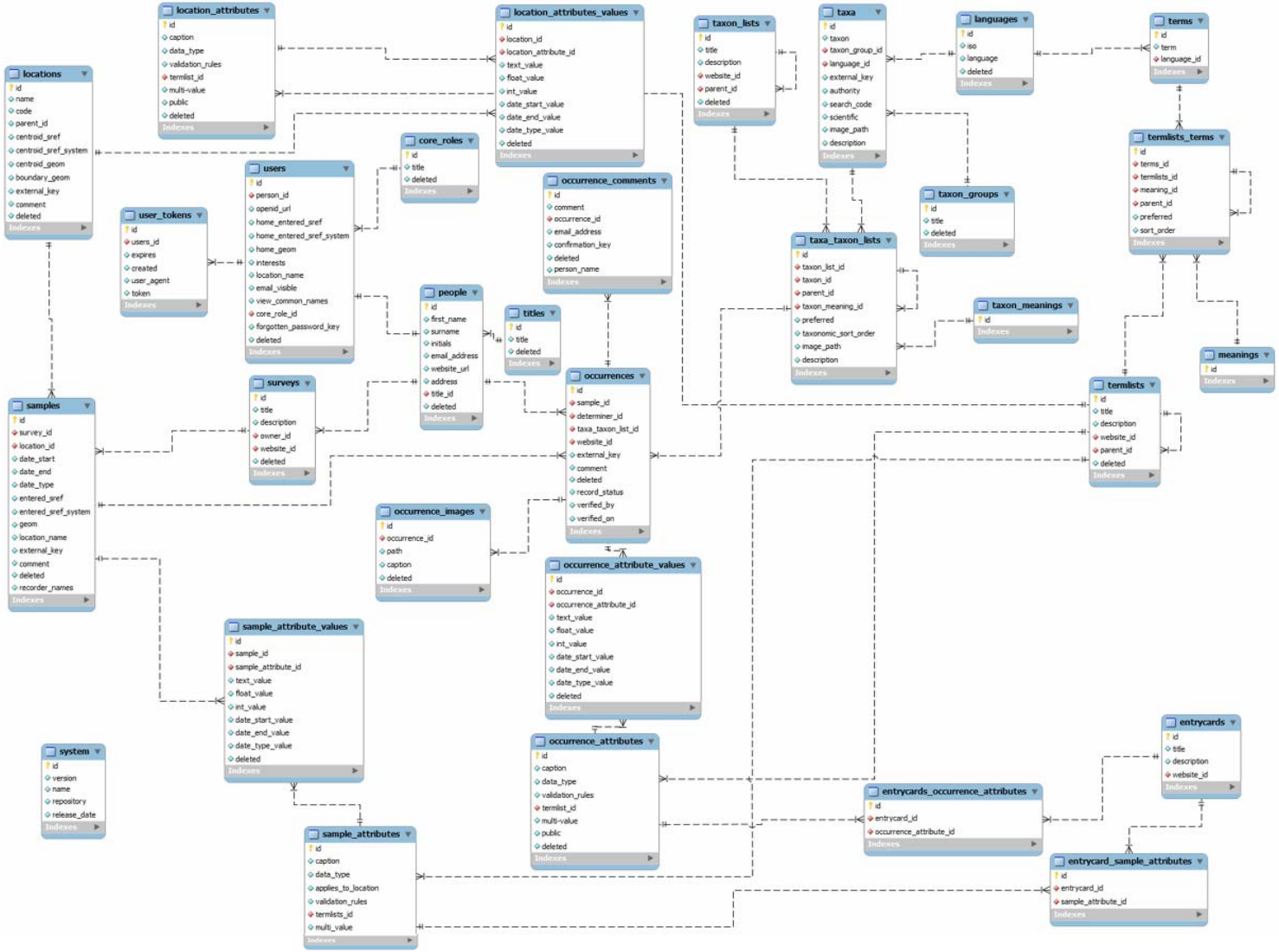
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Visualization algorithm



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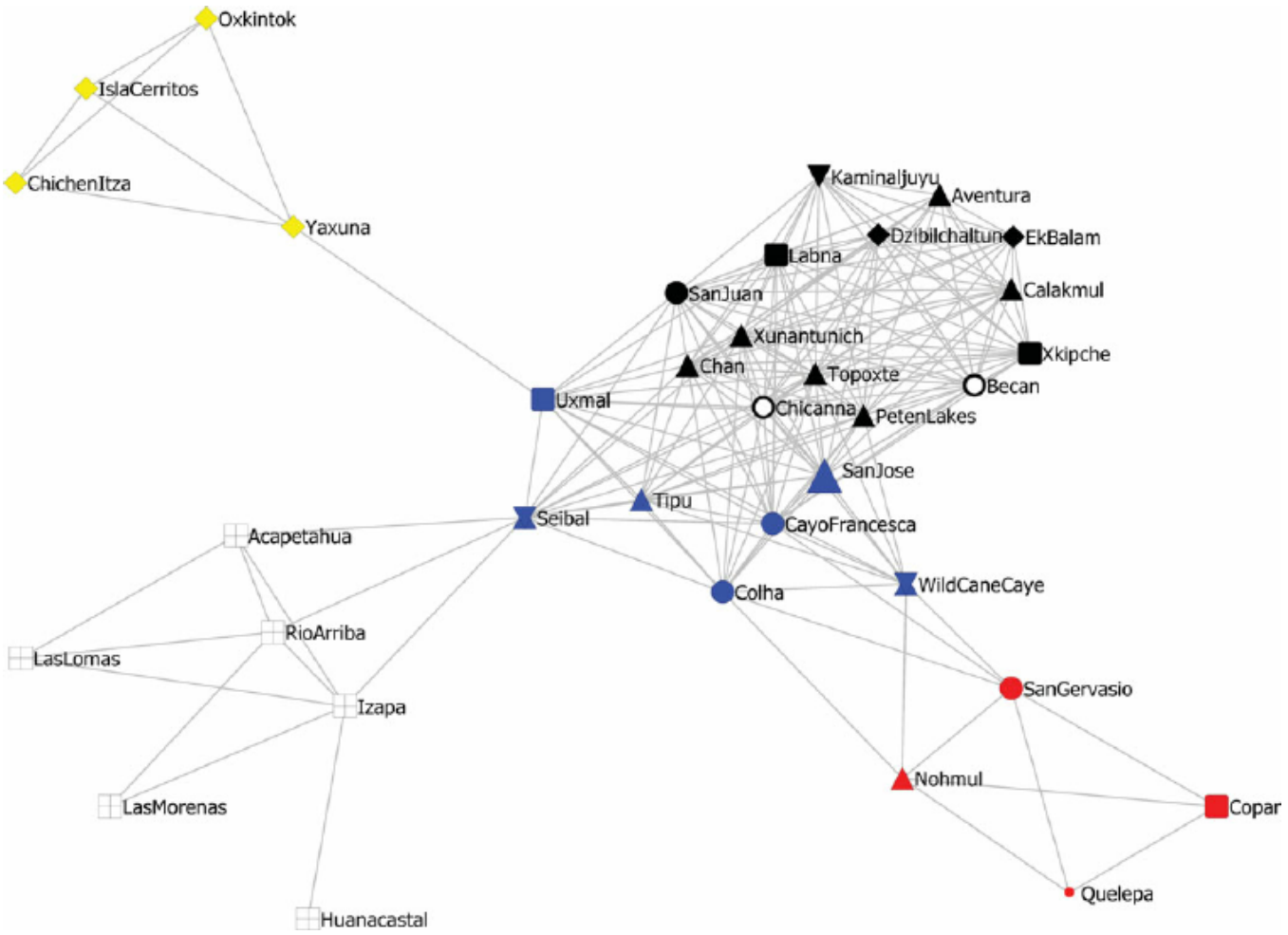
Example: a relational schema



- Orthogonal representation
- Aesthetic criteria
  - reduce crossings
  - reduce bends along edges
  - reduce area
  - ...

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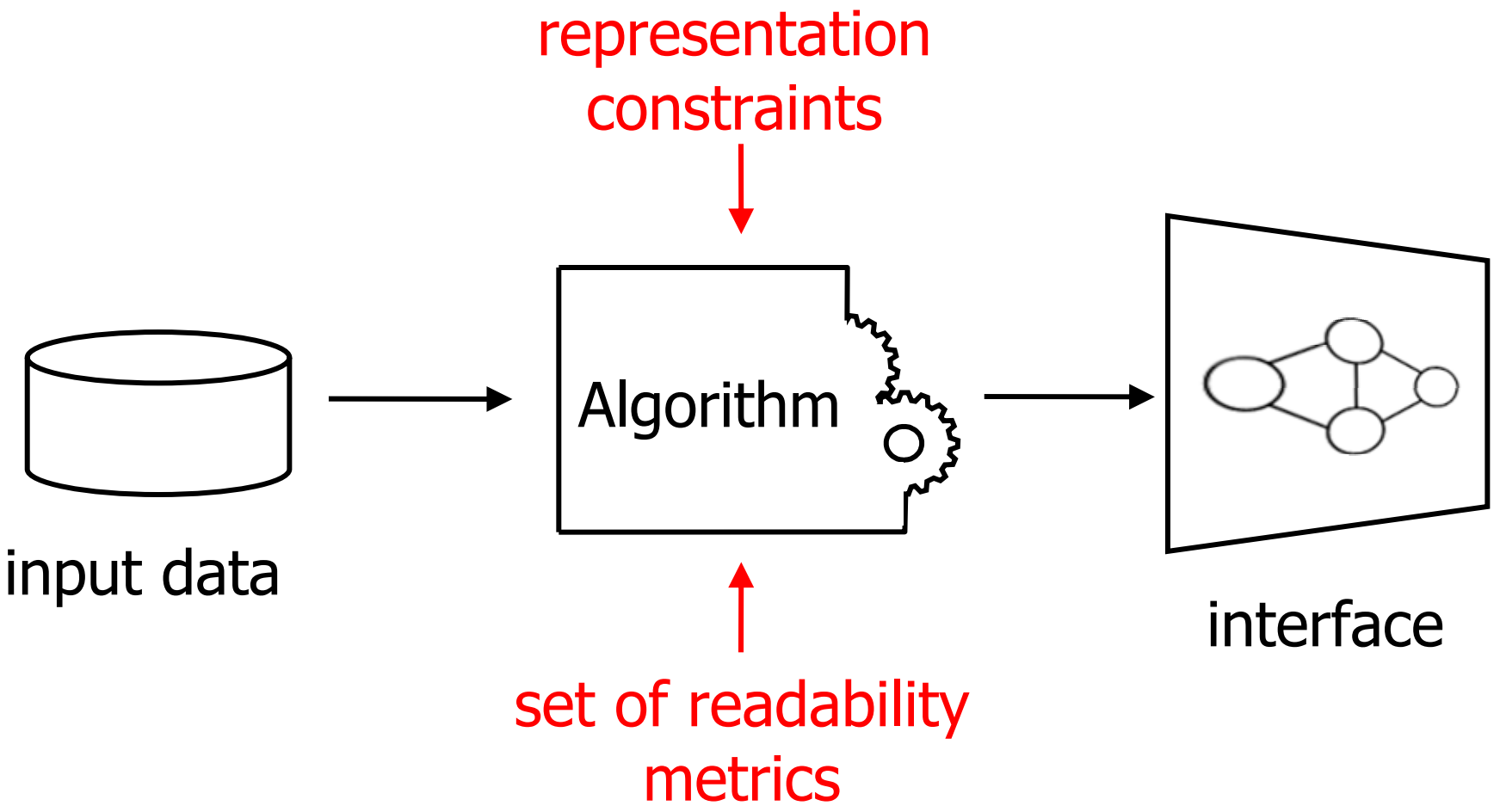
Example: relationships between Maya sites



- Straight-line representation
- Aesthetic criteria
  - reduce crossings
  - reduce area
  - increase angular resolution
  - increase distance among vertices
  - …

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Visualization algorithm



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Algorithm design and engineering

- Device efficient visualization algorithms to compute the needed representations
  - complying with the chosen representation conventions
  - taking into account the user preferences
- Devise efficient interaction algorithms to perform the operations that are triggered by the user
  - filtering, computing values, etc.
- The output is a description of the algorithmic steps

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Architectural and technological choices

- Purpose of architectural choices
  - decompose the system into major components
  - allocate functional responsibilities to components
  - define component interfaces
  - define communication protocols and interaction between components
- Purpose of technological choices
  - identify technologies and libraries able to support the needed graphical representations and user interactions

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Development, test, and debugging

- Development
  - implement each software component
  - integrate libraries through their API
- Test and debugging
  - identify functional or efficiency problems
  - isolate the source of the problem
  - correct the problem or determine a way to work around it

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Evaluation

- Measure the system performances with a scientifically sound method
  - measure its efficiency
    - is it quick enough to cope with user interactions
  - measure its effectiveness
    - is it better than previous contributions?
    - is the system effective and useful?
- Effectiveness assessments are hard to achieve because of the human in the loop

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Deployment

- Software deployment
  - decide how software is delivered to customer
  - provide documentation about how to install and use
- Service deployment
  - deliver the application over internet simply as a service
  - allow data to be accessed from any device using the Web

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Bibliography and further readings

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